

Asymptotic expansion of the state density function via the V -filtration

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Our aim in this note is to recast the key technical results in Chapter 4 of Watanabe's book [6] in the language of $\mathcal{D}$ -module theory, as far as we are able.

Throughout, let X denote a smooth real analytic variety of dimension d , and $f: X \rightarrow \mathbb{R}$ a nonnegative real analytic function.

1 Algebraic foundations

This section is mainly expository, following Popa's notes [4]. On a technical level, our main aim is to verify that the V -filtration construction works in the real analytic setting; due to its connection to monodromy, the V -filtration is normally only considered over $\mathbb{C}$.

1.1 Pushforward along a graph embedding

Let $Y = X \times \mathbb{R}$, with t the coordinate on the second factor, and let $\iota: X \hookrightarrow Y$ denote the graph embedding $x \mapsto (x, f(x))$. We consider the $\mathcal{D}$ -module pushforward $\iota_+ \mathcal{O}_X$. As an $\mathcal{O}_X$ -module, it can be explicitly described as

$$\iota_+ \mathcal{O}_X \cong \mathcal{O}_X \otimes_{\mathbb{R}} \mathbb{R}[\partial_t],$$

with left $\mathcal{D}_Y$ -action given by:

- (1) $g \cdot (h \otimes \partial_t^j) = (gh) \otimes \partial_t^j$, for $g \in \mathcal{O}_X$.
- (2) $t \cdot (h \otimes \partial_t^j) = (fh) \otimes \partial_t^j - jh \otimes \partial_t^{j-1}$.
- (3) $\partial_t \cdot (h \otimes \partial_t^j) = h \otimes \partial_t^{j+1}$.
- (4) $D \cdot (h \otimes \partial_t^j) = D(h) \otimes \partial_t^j - (D(f) \cdot h) \otimes \partial_t^{j+1}$, for $D \in \text{Der}(\mathcal{O}_X)$.

Rule (2) reflects the fact that $t = f$ on the image of ι : if $j = 0$, then $t \cdot (h \otimes 1) = (fh) \otimes 1$, consistent with $t|_{\iota(X)} = f$. The correction term $-jh \otimes \partial_t^{j-1}$ accounts for the commutation relation $[t, \partial_t] = -1$ in $\mathcal{D}_Y$.

There is a useful description of $\iota_+ \mathcal{O}_X$ in terms of the localisation $\mathcal{O}_X[t]_{f-t}$ of $\mathcal{O}_X[t]$ at the element $f-t$. The quotient $\mathcal{O}_X[t]_{f-t}/\mathcal{O}_X[t]$ has a natural $\mathcal{D}_Y$ -module structure, and is generated by the class

$$\delta_f := \left[\frac{1}{f-t} \right],$$

since every element can be written as a finite sum $\sum_{j \geq 0} h_j \partial_t^j \delta_f$ with $h_j \in \mathcal{O}_X$. This shows we have an $\mathcal{D}_Y$ -module isomorphism

$$\iota_+ \mathcal{O}_X \cong \mathcal{O}_X[t]_{f-t}/\mathcal{O}_X[t], \tag{1}$$

which identifies the generator $1 \otimes 1 \in \iota_+ \mathcal{O}_X$ with δ_f .

1.2 The Bernstein-Sato polynomial

The Bernstein–Sato polynomial (or b -function) of f is the key algebraic invariant underlying the V -filtration. To define it, we consider the $\mathcal{D}_X[s]$ -module $\mathcal{O}_X[\frac{1}{f}][s] \cdot f^s$, where f^s is a formal symbol and a derivation $D \in \text{Der}(\mathcal{O}_X)$ acts by

$$D(g \cdot f^s) = \left(D(g) + s g \frac{D(f)}{f} \right) f^s.$$

Theorem 1 (Bernstein–Sato). *There exist a polynomial $b(s) \in \mathbb{R}[s]$ and a differential operator $P(s) \in \mathcal{D}_X[s]$ such that*

$$P(s) \cdot f^{s+1} = b(s) \cdot f^s \quad (2)$$

holds in $\mathcal{O}_X[\frac{1}{f}][s] \cdot f^s$.

Proof. See Björk [2] (Theorem 3.2.6) for a proof in the analytic setting, which covers the real analytic case. $\square$

Definition 2. The *Bernstein–Sato polynomial* $b_f(s)$ is the monic polynomial of minimal degree satisfying (2).

Theorem 3 (Kashiwara). *The roots of $b_f(s)$ are negative rational numbers.*

Example 4. For $f = x^2 + y^2 + z^2$, we have $(\partial_x^2 + \partial_y^2 + \partial_z^2)(x^2 + y^2 + z^2)^{s+1} = (s+1)(4s+6)(x^2 + y^2 + z^2)^s$, so $b_f(s) = (s+1)(s+\frac{3}{2})$. The roots are -1 and $-\frac{3}{2}$.

1.3 The V -filtration

Let $s = -\partial_t t$. The V -filtration $\{V^\gamma\}_{\gamma \in \mathbb{Q}}$ is the unique rational filtration on $\iota_+ \mathcal{O}_X$ which is decreasing, exhaustive, discrete, and left-continuous, and which satisfies the following properties:

- (i) Each V^γ is a coherent module over $\mathcal{D}_X[t, s]$.
- (ii) $t \cdot V^\gamma \subset V^{\gamma+1}$, with equality for $\gamma > 0$.
- (iii) $\partial_t \cdot V^\gamma \subset V^{\gamma-1}$.
- (iv) The operator $s + \gamma$ acts nilpotently on $\text{gr}_V^\gamma := V^\gamma / V^{>\gamma}$, where $V^{>\gamma} := \cup_{\gamma' > \gamma} V^{\gamma'}$.

Uniqueness is a formal consequence of (i)–(iv). The proof of existence follows Malgrange’s construction as presented in [4], Theorem 3.2.6. We outline the argument and identify the points that require verification in the real analytic setting.

The original construction, for X a complex algebraic variety, has three steps:

- **Step 1.** One constructs a $\mathcal{D}_X[s]$ -submodule $N' \subset \iota_+ \mathcal{O}_X(*D)$ satisfying $t^k N_f \subset N' \subset t^m N_f$ (where $N_f = \mathcal{D}_X[s] f^s$ is the module generated by f^s), such that $tN' \subset N'$ and the action of $-s$ on N'/tN' has minimal polynomial with roots in $\mathbb{Q} \cap (0, 1]$. This uses the eigenspace decomposition for the action of s on $t^m N_f / t^k N_f$, which is algebraic, and the rationality of the roots of $b_f(s)$ (Kashiwara’s theorem).
- **Step 2.** From N' one defines a filtration W^α on $\iota_+ \mathcal{O}_X(*D)$ by integer shifts: $W^{\beta_i+q} = t^q W^{\beta_i}$ where $\beta_1 < \dots < \beta_\ell$ are the roots of the minimal polynomial from Step 1. The verification that $W^\bullet$ is a V -filtration is algebraic, using the commutation relation $P(s)t = tP(s-1)$ (equation (2.1.2) of [4]).
- **Step 3.** The V -filtration on $\iota_+ \mathcal{O}_X$ is defined by restriction: $V^\alpha \iota_+ \mathcal{O}_X = W^\alpha \cap \iota_+ \mathcal{O}_X$. This uses Lemma 3.2.7 of [4], which shows $N' \subset \iota_+ \mathcal{O}_X$.

Proposition 5. *The V -filtration on $\iota_+ \mathcal{O}_X$ exists and is unique in the real analytic category.*

Proof sketch. The key inputs to the construction are:

- (1) *Existence of the Bernstein–Sato polynomial.* This holds in the real analytic setting by Björk [2].
- (2) *Rationality of the roots of $b_f(s)$.* Since $b_f(s) \in \mathbb{Q}[s]$ (Kashiwara’s theorem, which holds for real analytic functions by the coincidence $b_f = b_{f_{\mathbb{C}}}$, see [5], §2.5), the roots are rational.
- (3) *Finite length of holonomic modules.* The proof of Step 1 uses the fact that $\mathcal{D}_{X_s} f^s$ is holonomic (over the base field $\mathbb{R}(s)$) and therefore has finite length, to conclude that the chain $\cdots \subset \mathcal{D}_{X_s} f^2 f^s \subset \mathcal{D}_{X_s} f f^s \subset \mathcal{D}_{X_s} f^s$ stabilizes. The holonomicity of $\mathcal{D}_{X_s} f^s$ follows from the same argument as in the complex case (the characteristic variety is contained in the zero section of T^*U_s), and the finite length property for holonomic $\mathcal{D}$ -modules holds in the real analytic setting by Björk [2] (Proposition 3.1.5).
- (4) *Algebraic steps.* Steps 2 and 3 of the construction, and the uniqueness proof, are purely algebraic manipulations (eigenspace decompositions, the commutation relation (2.1.2), and the intersection $W^\alpha \cap \iota_+ \mathcal{O}_X$). They apply unchanged.

□

We write $0 < \alpha_1 < \alpha_2 < \cdots$ for the *jumping indices* of the V -filtration (i.e. the values of γ at which $\text{gr}_V^\gamma \neq 0$), and m_k for the nilpotency order of $s + \alpha_k$ on $\text{gr}_V^{\alpha_k}$.

Corollary 6. *If $\alpha > 0$ is a jumping index of the V -filtration on $\iota_+ \mathcal{O}_X$ (i.e. $\text{gr}_V^\alpha \neq 0$), then α is an integer translate of λ for some root $-\lambda$ of $b_f(s)$.*

Proof. By construction (Step 1), the eigenvalues of $-s$ on N'/tN' are of the form β_i , each congruent to some λ modulo $\mathbb{Z}$ where $-\lambda$ is a root of $b_f(s)$. The V -filtration jumps only at rational numbers of the form $\beta_i + \mathbb{Z}$, giving the result. □

2 Asymptotic analysis of the state density function

In this section, we show that the generator $\delta_f \in \iota_+ \mathcal{O}_X$ has a distributional avatar in the form of the state density function $v(t)$, and use the V -filtration to derive its asymptotic expansion.

2.1 The distributional solution

Let $C_0^\infty(Y)$ denote the space of test functions on Y , and let $\mathcal{D}'(Y)$ denote its continuous dual — the space of Schwartz distributions on Y . The space $\mathcal{D}'(Y)$ carries a left $\mathcal{D}_Y$ -module structure defined by

$$\langle P \cdot u, \Phi \rangle := \langle u, P^t \Phi \rangle \quad \text{for } P \in \mathcal{D}_Y, u \in \mathcal{D}'(Y), \Phi \in C_0^\infty(Y),$$

where P^t is the formal transpose: $g^t = g$ for multiplication operators, $\partial_{x_i}^t = -\partial_{x_i}$, $\partial_t^t = -\partial_t$, and $(PQ)^t = Q^t P^t$.

Definition 7. The distribution $\Delta_f \in \mathcal{D}'(Y)$ is defined by

$$\langle \Delta_f, \Phi \rangle := \int_X \Phi(x, f(x)) dx \quad \text{for } \Phi \in C_0^\infty(Y).$$

In the usual notation, $\Delta_f = \delta(t - f(x))$: it is supported on the graph $\Gamma_f = \{(x, t) : t = f(x)\}$ and integrates test functions over it.

Proposition 8. *There is a unique $\mathcal{D}_Y$ -module homomorphism*

$$\sigma: \iota_+ \mathcal{O}_X \longrightarrow \mathcal{D}'(Y)$$

satisfying $\sigma(\delta_f) = \Delta_f$. It is given by $\sigma(h\partial_t^j \delta_f) = h(x) \cdot \partial_t^j \Delta_f$ for all $h \in \mathcal{O}_X$ and $j \geq 0$.

Proof. Uniqueness. Since δ_f generates $\iota_+ \mathcal{O}_X$ as a $\mathcal{D}_Y$ -module (Section 1.2), any $\mathcal{D}_Y$ -homomorphism is determined by $\sigma(\delta_f)$. The formula $\sigma(h\partial_t^j \delta_f) = h\partial_t^j \Delta_f$ then follows from the $\mathcal{D}_Y$ -linearity and the action rules (1)–(4).

Existence. We must verify that Δ_f satisfies the defining relations of $\iota_+ \mathcal{O}_X$, i.e., that σ respects the $\mathcal{D}_Y$ -action. Rules (1) and (3) (action of $g \in \mathcal{O}_X$ and ∂_t) are immediate from the definitions. The two nontrivial relations are:

Action of t (Rule 2). We must show $t \cdot \partial_t^j \Delta_f = f \partial_t^j \Delta_f - j \partial_t^{j-1} \Delta_f$. Pairing the left side with $\Phi \in C_0^\infty(Y)$:

$$\langle t \cdot \partial_t^j \Delta_f, \Phi \rangle = (-1)^j \int_X \partial_t^j (t\Phi) \big|_{t=f(x)} dx.$$

By the Leibniz rule, $\partial_t^j (t\Phi) = t \partial_t^j \Phi + j \partial_t^{j-1} \Phi$. Evaluating at $t = f(x)$ and comparing with the right side (where $f(x)$ and $\partial_{x_i} f$ are independent of t , so pass through ∂_t^j) gives the result.

Action of ∂_{x_i} (Rule 4). We must show $\partial_{x_i} \cdot \partial_t^j \Delta_f = -(\partial_{x_i} f) \partial_t^{j+1} \Delta_f$. Pairing the left side with Φ :

$$\langle \partial_{x_i} \partial_t^j \Delta_f, \Phi \rangle = (-1)^{j+1} \int_X (\partial_t^j \partial_{x_i} \Phi)(x, f(x)) dx.$$

Pairing the right side:

$$\langle -(\partial_{x_i} f) \partial_t^{j+1} \Delta_f, \Phi \rangle = (-1)^j \int_X (\partial_{x_i} f)(x) (\partial_t^{j+1} \Phi)(x, f(x)) dx.$$

These are equal by the chain rule identity

$$\frac{\partial}{\partial x_i} [(\partial_t^j \Phi)(x, f(x))] = (\partial_{x_i} \partial_t^j \Phi)(x, f(x)) + (\partial_{x_i} f)(x) (\partial_t^{j+1} \Phi)(x, f(x)),$$

integrated over X : the left side vanishes (since Φ has compact support, the composite $w \mapsto (\partial_t^j \Phi)(x, f(x))$ is compactly supported, and the integral of a total derivative of a compactly supported function is zero), giving the required identity. $\square$

2.2 The state density function

Note that pairing Δ_f with a test function of the form $\Phi(x, t) = \phi(x)\psi(t)$ gives $\int_X \phi(x)\psi(f(x)) dx$. This leads to the following definition.

Definition 9. Let $\phi \in C_0^\infty(X)$ with $\phi \geq 0$. The *state density distribution* associated to f and ϕ is the distribution $v \in \mathcal{D}'(\mathbb{R})$ defined by

$$\int_{\mathbb{R}} \psi(t) v(t) dt := \int_X \psi(f(x)) \phi(x) dx \quad \text{for all } \psi \in C_0^\infty(\mathbb{R}). \quad (3)$$

Equivalently, v is the pushforward of the measure $\phi(w) dx$ under $f: X \rightarrow \mathbb{R}$.

We write $v(t) = \int_X \delta(t - f(x)) \phi(x) dx$ as a formal shorthand for (3).

Proposition 10. *The distribution v is represented by a locally integrable function on $(0, \infty)$, smooth away from the critical values of f . At a regular value $t_0 > 0$ of f , it is given by the coarea formula:*

$$v(t_0) = \int_{f^{-1}(t_0)} \frac{\phi(x)}{|\nabla f(x)|} dS,$$

where dS is the surface measure on the level set $f^{-1}(t_0)$.

Proof. See Watanabe [6], Theorem 4.3. □

The function $v(t)$ is called the *state density function*. Note that since f is real analytic and nonnegative, the set of critical values of f is discrete in $[0, \infty)$, by the analytic Sard theorem. In particular, there exists $\epsilon > 0$ such that 0 is the only critical value in $[0, \epsilon)$, and $v(t)$ is smooth on $(0, \epsilon)$. More generally, we have the following result, which we will need for the asymptotic analysis:

Proposition 11. *For any $u \in \iota_+ \mathcal{O}_X$ and any $\phi \in C_0^\infty(X)$, the distribution $\langle u, \phi \rangle$ is represented by a smooth function on $(0, \epsilon)$.*

Proof. Every element $u \in \iota_+ \mathcal{O}_X$ is a finite sum $u = \sum_j h_j \partial_t^j \delta_f$ with $h_j \in \mathcal{O}_X$. By the distributional solution map (Proposition 8),

$$\langle u, \phi \rangle = \sum_j \partial_t^j v_{h_j \phi},$$

where $v_{h_j \phi}(t) = \int_X \delta(t - f(x)) h_j(x) \phi(x) dx$ is the state density with test function $h_j \phi \in C_0^\infty(X)$. By Proposition 10, each $v_{h_j \phi}$ is smooth on $(0, \epsilon)$, and therefore so is each $\partial_t^j v_{h_j \phi}$. Since the sum is finite, $\langle u, \phi \rangle$ is smooth on $(0, \epsilon)$. □

Remark 12. This proposition ensures that the asymptotic statements in the regularity theorem (Theorem 15) and the asymptotic expansion (Theorem 16) concern smooth functions on $(0, \epsilon)$, and the big O estimates have their usual meaning.

For any $u \in \iota_+ \mathcal{O}_X$ and $\phi \in C_0^\infty(X)$, define the distribution $\langle u, \phi \rangle \in \mathcal{D}'(\mathbb{R})$ by

$$\int_{\mathbb{R}} \psi(t) \langle u, \phi \rangle(t) dt := \langle \sigma(u), \phi(w) \psi(t) \rangle \quad \text{for all } \psi \in C_0^\infty(\mathbb{R}).$$

In particular, $\langle \delta_f, \phi \rangle = v(t)$. By Proposition 11, $\langle u, \phi \rangle$ is representable by a smooth function locally near zero. As smooth functions, the $\mathcal{D}_Y$ -linearity of σ gives:

- (a) $\langle t \cdot u, \phi \rangle = t \cdot \langle u, \phi \rangle$.
- (b) $\langle s \cdot u, \phi \rangle = -\partial_t t \cdot \langle u, \phi \rangle$, where $s = -\partial_t t$.

Property (a) holds because $(t \cdot \sigma(u))$ paired with $\phi(w) \psi(t)$ gives $\sigma(u)$ paired with $\phi(w)(t\psi(t))$, which equals $t \cdot \langle u, \phi \rangle$ paired with ψ . Property (b) follows similarly from the $\mathcal{D}_Y$ -linearity of σ and the transpose formula $\partial_t^t = -\partial_t$.

2.3 Asymptotics of the V -filtration

The distributional solution allows us to translate the algebraic structure of the V -filtration into analysis. Recall that with $s = -\partial_t t$, the operator $s + \alpha$ acts nilpotently on the associated graded piece gr_V^α . Now the homogeneous solutions of the differential equation $(-\partial_t t + \alpha)^m w = 0$ are $\mathbb{R}$ -linear combinations of the functions $t^{\alpha-1}(-\log t)^j$, for $0 \leq j \leq m-1$. To apply this observation to the V -filtration, we need the following asymptotic generalisation:

Lemma 13 (Inhomogeneous regular singular ODE). *Let $\alpha > 0$, $\beta > \alpha$, and $m \geq 1$ be a positive integer. Suppose $g: (0, \epsilon) \rightarrow \mathbb{R}$ satisfies*

$$g(t) = O(t^{\beta-1} |\log t|^M)$$

as $t \rightarrow 0^+$, for some integer $M \geq 0$. If $w: (0, \epsilon) \rightarrow \mathbb{R}$ is a smooth solution of

$$(-\partial_t t + \alpha)^m w = g,$$

then there exist constants $c_0, c_1, \dots, c_{m-1} \in \mathbb{R}$ such that

$$w(t) = \sum_{j=0}^{m-1} c_j t^{\alpha-1} (-\log t)^j + O(t^{\beta-1} |\log t|^{M+m-1})$$

as $t \rightarrow 0^+$.

Proof. We proceed by induction on m . Writing out the operator, $(-\partial_t t + \alpha)w = -tw' - w + \alpha w = -tw' + (\alpha - 1)w$.

Base case ($m = 1$). We solve

$$-tw'(t) + (\alpha - 1)w(t) = g(t).$$

The homogeneous solutions are $ct^{\alpha-1}$. Substituting $w(t) = t^{\alpha-1}h(t)$, we obtain

$$-t^\alpha h'(t) = g(t), \quad \text{i.e.} \quad h'(t) = -t^{-\alpha} g(t).$$

Since $g(t) = O(t^{\beta-1} |\log t|^M)$ and $\beta > \alpha$, the function $t^{-\alpha} g(t) = O(t^{\beta-\alpha-1} |\log t|^M)$ is integrable near $t = 0$ (as $\beta - \alpha > 0$). Define the particular solution by

$$h(t) = \int_0^t s^{-\alpha} g(s) ds.$$

Then $h(t) = O(t^{\beta-\alpha} |\log t|^M)$, so $w_p(t) = t^{\alpha-1} h(t) = O(t^{\beta-1} |\log t|^M)$. The general solution is $w = ct^{\alpha-1} + w_p$, which has the required form.

Inductive step. Assume the result holds for $m - 1$. Factor the equation as

$$(-\partial_t t + \alpha)[(-\partial_t t + \alpha)^{m-1} w] = g.$$

Set $w_1 = (-\partial_t t + \alpha)^{m-1} w$. By the base case, there exists a constant c_{m-1} such that

$$w_1(t) = c_{m-1} t^{\alpha-1} + O(t^{\beta-1} |\log t|^M).$$

Here $c_{m-1} t^{\alpha-1}$ is the homogeneous part (the free constant from the general solution of the first-order equation), and the $O(t^{\beta-1} |\log t|^M)$ term is the particular solution $w_p = t^{\alpha-1} \int_0^t s^{-\alpha} g(s) ds$.

Now w solves $(-\partial_t t + \alpha)^{m-1} w = w_1$. We apply the inductive hypothesis to this equation. The inhomogeneity is $w_1 = c_{m-1} t^{\alpha-1} + O(t^{\beta-1} |\log t|^M)$, which is $O(t^{\alpha-1})$. Since the inductive hypothesis requires $g = O(t^{\beta'-1} |\log t|^{M'})$ with $\beta' > \alpha$, it does not apply directly to w_1 when $c_{m-1} \neq 0$.

Instead, we first construct $w^{(\text{rem})}$ to be the specific particular solution of $(-\partial_t t + \alpha)^{m-1} w^{(\text{rem})} = w_p$, where $w_p = w_1 - c_{m-1} t^{\alpha-1} = O(t^{\beta-1} |\log t|^M)$ is the particular solution from the base case. Since $w_p = O(t^{\beta-1} |\log t|^M)$ with $\beta > \alpha$, the inductive hypothesis applies and gives

$$w^{(\text{rem})}(t) = \sum_{j=0}^{m-2} c'_j t^{\alpha-1} (-\log t)^j + O(t^{\beta-1} |\log t|^{M+m-2}).$$

Now set $w^{(\text{res})} := w - w^{(\text{rem})}$. Then $w^{(\text{res})}$ solves

$$(-\partial_t t + \alpha)^{m-1} w^{(\text{res})} = w_1 - w_p = c_{m-1} t^{\alpha-1}.$$

To analyse $w^{(\text{res})}$, we peel off one operator: set $w_2 = (-\partial_t t + \alpha)^{m-2} w^{(\text{res})}$, so that $(-\partial_t t + \alpha)w_2 = c_{m-1} t^{\alpha-1}$. Writing this out, $-tw'_2 + (\alpha - 1)w_2 = c_{m-1} t^{\alpha-1}$. Setting $w_2 = t^{\alpha-1} h$ gives

$-t^\alpha h' = c_{m-1} t^{\alpha-1}$, i.e. $h' = -c_{m-1} t^{-1}$. Define $h(t) = -c_{m-1} \log t + c'$ for an arbitrary constant c' . Then

$$w_2(t) = c_{m-1} t^{\alpha-1} (-\log t) + c' t^{\alpha-1},$$

which is a linear combination of $t^{\alpha-1} (-\log t)^j$ with $j \leq 1$. By iterating this argument (or applying the inductive hypothesis with g replaced by w_2 and β replaced by any value greater than α), we obtain

$$w^{(\text{res})}(t) = \sum_{j=0}^{m-1} c_j'' t^{\alpha-1} (-\log t)^j.$$

Combining the two parts and relabelling constants, we obtain

$$w(t) = \sum_{j=0}^{m-1} c_j t^{\alpha-1} (-\log t)^j + O(t^{\beta-1} |\log t|^{M+m-1}). \quad \square$$

Corollary 14 (Right inverse). *For $\beta > \alpha > 0$ and $M \geq 0$, the operator $(-\partial_t + \alpha)^m$ has a right inverse on the space of smooth functions on $(0, \epsilon)$ that are $O(t^{\beta-1} |\log t|^M)$: there is a linear map $g \mapsto R_\alpha g$ such that*

$$(-\partial_t + \alpha)^m (R_\alpha g) = g, \quad R_\alpha g = O(t^{\beta-1} |\log t|^{M+m-1}).$$

Proof. $R_\alpha g$ is the particular solution from Lemma 13 with all homogeneous constants set to zero. For $m = 1$, it is given explicitly by the integral $R_\alpha g(t) = t^{\alpha-1} \int_0^t s^{-\alpha} g(s) ds$, which converges at $s = 0$ since $\beta > \alpha$, and satisfies $R_\alpha g = O(t^{\beta-1} |\log t|^M)$. The general case $m > 1$ follows by iteration. $\square$

Theorem 15 (Asymptotics of the V -filtration). *Let $u \in V^\alpha$ with $\alpha > 0$. Then there exists an integer $N \geq 0$ such that*

$$\langle u, \phi \rangle = O(t^{\alpha-1} |\log t|^N)$$

as $t \rightarrow 0^+$.

Proof. By Corollary 6, every jumping number α_k of the V -filtration is of the form $\alpha_k = \lambda + n$ for some root $-\lambda$ of the Bernstein–Sato polynomial $b_f(s)$ and some $n \in \mathbb{Z}$. Since the roots of $b_f(s)$ are finitely many negative rationals, the full set of jumping numbers is contained in finitely many arithmetic progressions with common difference 1. We define

$$\beta_1 < \beta_2 < \dots < \beta_r$$

to be the fractional parts $\{-\lambda\} \in [0, 1)$ as λ ranges over the roots of $b_f(s)$. Not every β_i need be an actual jumping number of the V -filtration; those that are not will simply contribute trivially to the expansion. This allows us to organise the proof of the growth estimate in two levels: integer shifts (handled by multiplication by t) and the finitely many base cases in $[0, 1)$ (handled by the ODE estimate).

From here, the proof proceeds in three steps.

Step 1: A priori bound at V^1 . Since the first jumping number satisfies $\alpha_1 > 0$, we have $V^0 = V^{\alpha_1} = \iota_+ \mathcal{O}_X$, and $\delta_f \in V^0$. The state density function $v(t) = \langle \delta_f, \phi \rangle$ is locally integrable near $t = 0$, so $v(t) = O(t^{-1+\varepsilon})$ for every $\varepsilon > 0$. Since $t \cdot V^\gamma = V^{\gamma+1}$ for $\gamma > 0$, we have $t \cdot \delta_f \in V^1$, and

$$\langle t \cdot \delta_f, \phi \rangle = t v(t) = O(t^\varepsilon)$$

for every $\varepsilon > 0$. Since the jumping numbers are discrete and $\beta_r < 1$, there exists a rational $\delta > 0$ such that $V^{1-\delta} = V^1$ (i.e. there are no jumping numbers in $(1 - \delta, 1)$). Taking $\varepsilon = \delta$, the a priori bound gives: for any $u \in V^{1-\delta}$, $\langle u, \phi \rangle = O(t^\delta) = O(t^{(1-\delta)-1})$. This seeds the downward induction in Step 2.

Step 2: Base cases in $[0, 1)$ (ODE estimate). Let $\beta_1 < \beta_2 < \dots < \beta_r$ be the fractional parts defined above; note $\beta_r < 1$. We establish the sharp estimate at each β_i by finite downward induction on $i = r, r-1, \dots, 1$.

Let $u \in V^{\beta_i}$. If β_i is not an actual jumping number of the V -filtration, then $\text{gr}_V^{\beta_i} = 0$, so $V^{\beta_i} = V^{>\beta_i} \subset V^{\beta_{i+1}}$ and there is nothing to prove at this level. Otherwise, the nilpotency of $s + \beta_i$ on $\text{gr}_V^{\beta_i}$ gives

$$(s + \beta_i)^{m_i} u \in V^{>\beta_i} \subset V^{\beta_{i+1}},$$

where we set $\beta_{r+1} := 1 - \delta$ (with δ as in Step 1). By the intertwining property, $w(t) := \langle u, \phi \rangle$ satisfies the ODE

$$(-\partial_t + \beta_i)^{m_i} w(t) = g(t),$$

where $g(t) = \langle (s + \beta_i)^{m_i} u, \phi \rangle$. For $i = r$, the a priori bound from Step 1 gives $g(t) = O(t^{(1-\delta)-1})$. Since $\beta_r < 1 - \delta$, Lemma 13 applies with $\beta = 1 - \delta$ and $M = 0$, yielding

$$w(t) = \sum_{j=0}^{m_r-1} c_j t^{\beta_r-1} (-\log t)^j + O(t^{(1-\delta)-1} |\log t|^{m_r-1}).$$

This establishes the estimate at β_r and simultaneously provides a sharp bound at depth $\beta_{r+1} = 1 - \delta$, which seeds the estimate for β_{r-1} . Continuing downward through $\beta_{r-1}, \dots, \beta_1$ completes the base cases.

Step 3: Integer shifts (multiplication by t). For $\gamma > 0$, the equality $V^{\gamma+1} = t \cdot V^\gamma$ means that every $u \in V^{\gamma+1}$ can be written as $u = t \cdot w$ with $w \in V^\gamma$. By the intertwining property,

$$\langle u, \phi \rangle = t \cdot \langle w, \phi \rangle.$$

If $\langle w, \phi \rangle = O(t^{\gamma-1} |\log t|^N)$, then $\langle u, \phi \rangle = O(t^\gamma |\log t|^N) = O(t^{(\gamma+1)-1} |\log t|^N)$.

Since every jumping number $\alpha_k > 1$ is of the form $\beta_i + n$ for some $\beta_i \in [0, 1)$ and $n \geq 1$ (where β_i is one of the fractional parts defined above), the estimate at α_k follows from the estimate at β_i by n applications of this shift. $\square$

Finally, we can use these results to obtain the asymptotic expansion of $v(t)$.

Theorem 16. *The state density function $v(t) = \langle \delta_f, \phi \rangle$ admits an asymptotic expansion as $t \rightarrow 0^+$ of the form*

$$v(t) \sim \sum_{k=1}^{\infty} \sum_{m=1}^{m_k} D_{km}(\phi) t^{\alpha_k-1} (-\log t)^{m-1},$$

where $0 < \alpha_1 < \alpha_2 < \dots$ are the jumping numbers of the V -filtration on $\iota_* \mathcal{O}_X$, m_k is the nilpotency order of $s + \alpha_k$ on $\text{gr}_V^{\alpha_k}$, and D_{km} are Schwartz distributions depending on the V -filtration.

Proof. Define $u_k = \prod_{j=1}^k (s + \alpha_j)^{m_j} \delta_f$ for $k \geq 0$ (with $u_0 = \delta_f$). Since $(s + \alpha_k)^{m_k}$ is nilpotent on $\text{gr}_V^{\alpha_k}$, we have $u_k \in V^{\alpha_{k+1}}$. Set $w_k = \langle u_{k-1}, \phi \rangle$ and $P_k = (-\partial_t + \alpha_k)^{m_k}$. By the intertwining property, $P_k w_k = w_{k+1}$, and by Theorem 15, $w_{k+1} = O(t^{\alpha_{k+1}-1} |\log t|^{N_k})$. In particular $w_1 = v(t)$, and composing:

$$P_N \circ \dots \circ P_1 v = w_{N+1} = O(t^{\alpha_{N+1}-1} |\log t|^M).$$

Particular solution. We construct a particular solution v_p with $v_p = O(t^{\alpha_{N+1}-1} |\log t|^{M'})$ by solving the factored system from the outside in. Set $f_{N+1} = w_{N+1}$. For $k = N, N-1, \dots, 1$, let $f_k = R_{\alpha_k} f_{k+1}$ be the right inverse from Corollary 14. At each step, $f_{k+1} = O(t^{\alpha_{N+1}-1} |\log t|^{M_k})$ and $\alpha_{N+1} > \alpha_k$, so the corollary applies and gives $f_k = O(t^{\alpha_{N+1}-1} |\log t|^{M_{k-1}})$: the growth exponent $\alpha_{N+1} - 1$ is preserved, and only the log power increases. Then $v_p := f_1$ satisfies $P_N \circ \dots \circ P_1 v_p = w_{N+1}$ and $v_p = O(t^{\alpha_{N+1}-1} |\log t|^{M'})$.

Homogeneous part. The difference $v - v_p$ satisfies $P_N \circ \dots \circ P_1 (v - v_p) = 0$. Since $\alpha_1 < \dots < \alpha_N$ are distinct, the homogeneous solutions are $\{t^{\alpha_k-1}(-\log t)^j : k = 1, \dots, N, j = 0, \dots, m_k - 1\}$, so

$$v(t) = \sum_{k=1}^N \sum_{j=0}^{m_k-1} c_{kj} t^{\alpha_k-1} (-\log t)^j + O(t^{\alpha_{N+1}-1} |\log t|^{M'}).$$

The coefficients c_{kj} are uniquely determined, since asymptotic expansions with respect to the scale $\{t^{\gamma-1}(-\log t)^j : \gamma \in \mathbb{Q}_{>0}, j \in \mathbb{Z}_{\geq 0}\}$ are unique. Setting $D_{km}(\phi) := c_{k,m-1}$ gives the expansion. $\square$

Lemma 17. *For each jumping index α_k and each $m = 1, \dots, m_k$, the map $\phi \mapsto D_{km}(\phi)$ is a distribution on X of finite order.*

Proof. Linearity is clear: $v(t)$ depends linearly on ϕ , v_p depends linearly on ϕ (since $w_{N+1} = \langle u_N, \phi \rangle$ is linear in ϕ and each right inverse R_{α_k} is linear), so $v - v_p$ is linear in ϕ , and hence its coefficients c_{kj} in the basis $\{t^{\alpha_k-1}(-\log t)^j\}$ are linear in ϕ .

For continuity, we show that $\phi \mapsto v(t) - v_p(t)$ is continuous from $C_0^\infty(X)$ into the space of smooth functions on $(0, \epsilon)$, and that the coefficient extraction is continuous.

The state density $v(t) = \int_{f^{-1}(t)} \phi / |\nabla f| dS$ (by Proposition 10) depends continuously on ϕ in the C^0 topology: $|v(t)| \leq \|\phi\|_{C^0} \int_{f^{-1}(t)} dS / |\nabla f|$.

For v_p , the starting point $w_{N+1} = \langle u_N, \phi \rangle$ depends on ϕ through the distributional solution map. Since $u_N = \sum_\ell h_\ell \partial_t^\ell \delta_f$ (a fixed finite sum with $h_\ell \in \mathcal{O}_X$), we have $w_{N+1}(t) = \sum_\ell (-1)^\ell \partial_t^\ell v_{h_\ell \phi}(t)$, so $|w_{N+1}(t)| \leq C \|\phi\|_{C^L}$ for some fixed L depending on the orders ℓ appearing. Each right inverse $R_{\alpha_k} g(t) = t^{\alpha_k-1} \int_0^t s^{-\alpha_k} g(s) ds$ (for $m_k = 1$; iterated for $m_k > 1$) preserves the bound: $|R_{\alpha_k} g(t)| \leq C' t^{\beta-1} \|g\|$ where $\|g\|$ is the sup of $|s^{1-\beta} g(s)|$. Composing N such operators, $|v_p(t)| \leq C'' \|\phi\|_{C^L} t^{\alpha_{N+1}-1}$.

Therefore $v(t) - v_p(t)$ depends continuously on ϕ in the C^L topology for each t , and the coefficient $c_{kj} = D_{k,j+1}(\phi)$ — being the unique coefficient of $t^{\alpha_k-1}(-\log t)^j$ in the finite linear combination $v - v_p$ — is a continuous linear functional of ϕ , bounded by $C \|\phi\|_{C^L}$. Since $|D_{km}(\phi)| \leq C \|\phi\|_{C^L}$ and the $C_0^\infty(X)$ topology is finer than C^L , this shows D_{km} is continuous on $C_0^\infty(X)$, hence a distribution of order at most L . $\square$

3 The zeta function and real log canonical threshold

In this section we address the question: when do the terms of the asymptotic expansion in Theorem 16 vanish? Understanding this requires an alternative derivation of the expansion, which depends on the zeta function of f .

3.1 The zeta function

For $\phi \in C_0^\infty(X)$ with $\phi \geq 0$, define the *zeta function*

$$\zeta_\phi(z) = \int_X f(x)^z \phi(x) dx.$$

This converges absolutely for $\operatorname{Re}(z) > 0$ and defines a holomorphic function there. The analytic continuation to the entire complex plane is provided by the Bernstein–Sato polynomial $b_f(s)$, which satisfies the functional equation

$$D(z, w, \partial_w) f(x)^{z+1} = b_f(z) f(x)^z,$$

where D is a differential operator depending polynomially on z . Integrating by parts:

$$\zeta_\phi(z) = \frac{1}{b_f(z)} \int f(x)^{z+1} D^*(z, x, \partial_x) \phi(x) dx. \quad (4)$$

The right-hand side is a meromorphic function on the half-plane $\operatorname{Re}(z) > 1$, and its poles are the roots of $b_f(z)$ not cancelled by the vanishing of the integral. Iterating this argument, we may analytically continue $\zeta_\phi(z)$ the entire complex plane. The poles of $\zeta_\phi(z)$ are negative integer translates of the roots of $b_f(z)$, and in particular are rational numbers.

3.2 Asymptotic expansion of $\zeta(z)$ via the inverse Mellin transform

Watanabe [6], Section 4.5, derives the asymptotic expansion of $v(t)$ from the zeta function as follows. The zeta function is the Mellin transform of $v(t)$: $\zeta_\phi(z) = \int_0^\infty t^z v(t) dt$. If $\zeta_\phi(z)$ has poles at $0 > -\lambda_1 > -\lambda_2 > \dots$ with orders $\theta_1, \theta_2, \dots$, then the Laurent expansion of ζ_ϕ near each pole, combined with the inverse Mellin transform, gives

$$v(t) \sim \sum_{j=1}^{\infty} \sum_{m=1}^{\theta_j} D'_{jm}(\phi) t^{\lambda_j-1} (-\log t)^{m-1}$$

as $t \rightarrow 0^+$, where $D'_{jm}(\phi)$ are the Laurent coefficients (which are distributions in ϕ).

Proposition 18. *The V -filtration expansion (Theorem 16) and the Mellin expansion agree term by term.*

Proof. Both are asymptotic expansions of the same function $v(t)$ as $t \rightarrow 0^+$, with terms drawn from the asymptotic scale $\mathcal{S} = \{t^{\gamma-1}(-\log t)^j : \gamma \in \mathbb{Q}_{>0}, j \in \mathbb{Z}_{\geq 0}\}$. Since asymptotic expansions with respect to a countable scale are unique, the two agree term by term. $\square$

Since the V -filtration expansion has zero coefficients at non-jumping numbers, so the Mellin expansion does too, and hence ζ_ϕ has no pole at $-\gamma$ when $\gamma \in \mathbb{Q}$ is not a jumping index of the V -filtration.

Corollary 19. *We have the inclusion*

$$\bigcup_{\phi \in C_0^\infty(X)} \{-\lambda : \lambda \text{ is a pole of } \zeta_\phi\} \subset \{\text{jumping indices of the } V\text{-filtration on } \iota_+ \mathcal{O}_X\}.$$

We will show below that this inclusion is strict.

3.3 The real log canonical threshold

The index of the first nonvanishing term of the expansion of $v(t)$, known as the real log canonical threshold, is especially important in singular learning theory. In the analytic category, it is usually defined as follows:

Definition 20. The *real log canonical threshold* of f is the smallest $\alpha > 0$ such that $|f|^{-\alpha}$ is not locally integrable on X :

$$\operatorname{rlct}(f) = \inf\{\alpha > 0 : |f|^{-\alpha} \notin L_{\operatorname{loc}}^1(X)\}.$$

To connect this definition to the expansion of $v(t)$, let $\pi: X' \rightarrow X$ be a resolution of singularities such that $f \circ \pi$ is locally of the form $u \prod_i x_i^{m_i}$ where u is a nowhere-vanishing real analytic function, and the Jacobian $|\det d\pi|$ is locally $|u'| \prod_i x_i^{a_i}$ for u' nowhere vanishing. Let D'_j denote the irreducible components of the total transform with multiplicities m_j and Jacobian exponents a_j . Define

$$J_{\mathbb{R}} = \{j \in J : D'_j(\mathbb{R}) \neq \emptyset\}.$$

Proposition 21 (Geometric formula for the rlct). *The real log canonical threshold of f is*

$$\operatorname{rlct}(f) = \min_{j \in J_{\mathbb{R}}} \frac{a_j + 1}{m_j}.$$

Proof. By the change of variables formula, $\int |f|^{-\alpha} \phi dx = \sum_{\ell} \int |u|^{-\alpha} \prod_i x_i'^{a_i - \alpha m_i} \Phi_{\ell} dx'$ (via partition of unity on X'), where $\Phi_{\ell} \geq 0$ involves $\phi \circ \pi$ and $|u'|$. Each factor $\int_0^{\epsilon} x_i'^{a_i - \alpha m_i} dx_i'$ converges if and only if $a_i - \alpha m_i > -1$, i.e. $\alpha < (a_i + 1)/m_i$. The integral over the real locus only involves divisors $j \in J_{\mathbb{R}}$ (the others have empty real locus). So $|f|^{-\alpha}$ is locally integrable if and only if $\alpha < (a_j + 1)/m_j$ for all $j \in J_{\mathbb{R}}$, and the threshold is the minimum. $\square$

Using the same resolution of singularities, we can also make a connection to ζ_{ϕ} :

Proposition 22. *The real log canonical threshold $\lambda = \text{rlct}(f)$ is the negation of the largest pole of ζ_{ϕ} for generic ϕ (i.e. for ϕ with $\phi(x) \neq 0$ at a point where $|f|^{-\lambda}$ fails to be integrable). Equivalently, λ is the index of the first nonvanishing term in the asymptotic expansion of $v(t)$.*

Proof. Using the resolution π as in Proposition 21, the integral $\int |f|^{-\alpha} \phi dx$ transforms to a sum of integrals in normal crossing charts:

$$\int |f|^{-\alpha} \phi dx = \sum_{\ell} \int \prod_i x_i'^{a_i - \alpha m_i} \Phi_{\ell}(x') dx',$$

where $\Phi_{\ell} \geq 0$. Each factor $\int_0^{\epsilon} x_i'^{a_i - \alpha m_i} dx_i'$ converges if and only if $a_i - \alpha m_i > -1$. For $\alpha < \text{rlct}(f)$, all such integrals converge (by definition of the minimum). For $\alpha = \text{rlct}(f)$, there exists $j \in J_{\mathbb{R}}$ with $a_j - \alpha m_j = -1$, and the corresponding integral diverges logarithmically. Choosing ϕ supported near $\pi(D'_{j, \mathbb{R}})$ ensures $\Phi_{\ell} > 0$ on the relevant divisor, so the divergence is not cancelled. $\square$

We can now show that the inclusion of Corollary 19 is strict in general. Consider f as a complex analytic function on the complexification $X_{\mathbb{C}}$. If we define a zeta function $\zeta'(z) = \int_{X_{\mathbb{C}}} |f_{\mathbb{C}}|^{2z} \phi(x) dx$, then arguments similar to the above show that the greatest pole of $\zeta'(z)$ is equal to $-\text{lct}(f_{\mathbb{C}})$, where $\text{lct}(f_{\mathbb{C}})$ is the (complex) log canonical threshold. Hence, the V -filtration on $\iota_+ \mathcal{O}_{X, \mathbb{C}}$ jumps at $\alpha := \text{lct}(f_{\mathbb{C}})$. But the Bernstein-Sato polynomial of f and its complexification $f_{\mathbb{C}}$ are equal, and the complex V -filtration on $\iota_+ \mathcal{O}_{X, \mathbb{C}}$ is obtained from the real V -filtration on $\iota_+ \mathcal{O}_X$ by a simple extension of scalars. In particular, the real V -filtration must also jump at α .

Now in general, $\text{lct}(f_{\mathbb{C}}) \leq \text{rlct}(f)$, but the inequality is strict whenever $\text{rlct}(f) > 1$, since it is always the case that $\text{lct}(f_{\mathbb{C}}) \leq 1$ (since the lct is always equal to the negative of the largest root of $b_f(s)$ ([4], Theorem 2.7.2), and $(s+1)$ always divides $b_f(s)$). So the V -filtration jumps at $\alpha \leq 1$, but if $\text{rlct}(f) > 1$, then α is not a pole of $\zeta_{\phi}(z)$ for any ϕ . For more on the difference between the lct and rlct , see [5].

Example 23. Let $f(x, y, z) = x^2 + y^2 + z^2$ on $X = \mathbb{R}^3$. In Example 4, we computed that $b_f(s) = (s+1)(s+\frac{3}{2})$. So $\text{lct}(f_{\mathbb{C}}) = 1$. On the other hand, one can check using (4) that the simple pole of $b_f(z)^{-1}$ at $z = -1$ is cancelled by the vanishing of $D(-1)$. So $\zeta(z)$ has no pole there, and $\text{rlct}(f) > 1$. In fact, $\text{rlct}(f) = \frac{3}{2}$, since it defines a nonsingular statistical model in the sense of [6].

4 The partition function

In this section we derive the main asymptotic results for singular integrals (Watanabe [6], Theorems 4.7–4.9). As we did not find a way to leverage the $\mathcal{D}$ -module perspective on $v(t)$ for this purpose, our presentation is essentially a repackaging of Watanabe's original proofs.

4.1 Laplace asymptotics

The state density $v_\phi(t) = \langle \delta_f, \phi \rangle$ and the partition function are related by the Laplace transform:

$$\int_X e^{-nf(w)} \phi(w) dw = \int_0^\infty e^{-nt} v_\phi(t) dt.$$

The asymptotic expansion of $v_\phi(t)$ as $t \rightarrow 0^+$ (Theorem 16) therefore controls the large- n asymptotics via the standard substitution $u = nt$.

Theorem 24 (Laplace asymptotics). *Let $p \geq 0$, and assume ϕ does not vanish identically on $f^{-1}(0)$ (so that the leading coefficient $D_{1,\theta}(\phi) > 0$). Then there exist constants $a_1, a_2 > 0$ such that for $n > 1$,*

$$a_1 \frac{(\log n)^{\theta-1}}{n^{\lambda+p/2}} \leq \int_X e^{-nf} f^{p/2} \phi dw \leq a_2 \frac{(\log n)^{\theta-1}}{n^{\lambda+p/2}}.$$

Proof. The integral equals $\int_0^\infty e^{-nt} t^{p/2} v_\phi(t) dt$. Substituting $u = nt$ and using $v_\phi(u/n) \sim D_{1,\theta}(\phi) (u/n)^{\lambda-1} (\log(n/u))^{\theta-1}$:

$$\int_0^\infty e^{-nt} t^{p/2} v_\phi(t) dt \sim \frac{D_{1,\theta}(\phi) (\log n)^{\theta-1}}{n^{\lambda+p/2}} \Gamma(\lambda + p/2).$$

The two-sided bound follows from the positivity of $D_{1,\theta}(\phi)$. □

4.2 The perturbed partition function

Following Remark 4.4 of [6], the partition function of singular learning theory is

$$Z^p(n, \xi, \phi) = \int_X f^{p/2} \exp(-n\beta f + \sqrt{n}\beta f^{1/2} \xi) \phi dw, \quad (5)$$

where $p \geq 0$, $\beta > 0$, $\xi \in C^1(X)$, $\phi \in C_0^\infty(X)$ with $\phi \geq 0$, and f admits an analytic square root $f^{1/2}$. In the Laplace representation:

$$Z^p(n, \xi, \phi) = \int_0^\infty e^{-n\beta t} t^{p/2} v_{\phi_n}(t) dt, \quad (6)$$

where $\phi_n(w) = e^{\sqrt{n}\beta f(w)^{1/2} \xi(w)} \phi(w)$ is an n -dependent test function. The state density $v_{\phi_n}(t) = \langle \delta_f, \phi_n \rangle$ depends on n through ϕ_n , so (6) is not a standard Laplace transform of a fixed function.

4.3 Two-sided bounds

The crude bounds follow from the AM–GM inequality without any expansion.

Corollary 25 (Bounds on the partition function). *There exist constants $a_1, a_2 > 0$ (depending on f , β , p , but not on ξ , ϕ , or n) such that for all $n > 1$:*

$$a_1 \frac{(\log n)^{\theta-1}}{n^{\lambda+p/2}} e^{-\beta \|\xi\|^2/4} \min \phi \leq Z^p(n, \xi, \phi) \leq a_2 \frac{(\log n)^{\theta-1}}{n^{\lambda+p/2}} e^{\beta \|\xi\|^2/4} \|\phi\|.$$

Proof. By AM–GM, $|\sqrt{n}\beta f^{1/2} \xi| \leq \frac{1}{2}(n\beta f + \beta \|\xi\|^2/2)$, so

$$e^{-\beta \|\xi\|^2/4} e^{-3n\beta f/2} \leq e^{-n\beta f + \sqrt{n}\beta f^{1/2} \xi} \leq e^{\beta \|\xi\|^2/4} e^{-n\beta f/2}.$$

Multiplying by $f^{p/2} \phi$ and integrating, then applying Theorem 24, gives the result. □

4.4 Expansion of the perturbed state density

To obtain the sharp error bound, we expand $v_{\phi_n}(t)$ in powers of $\sqrt{n}$. Writing $\phi_n = e^{\sqrt{n}\beta f^{1/2}\xi}\phi$ and expanding the exponential:

$$v_{\phi_n}(t) = \sum_{j=0}^{\infty} \frac{(\sqrt{n}\beta)^j}{j!} \langle \delta_f, f^{j/2} \xi^j \phi \rangle = \sum_{j=0}^{\infty} \frac{(\sqrt{n}\beta)^j}{j!} t^{j/2} v_j(t),$$

where $v_j(t) = \langle \delta_f, \xi^j \phi \rangle$ is the state density with test function $\xi^j \phi$ (the factor $f^{j/2}$ becomes $t^{j/2}$ on the support of the delta function). The asymptotic expansion of each v_j (Theorem 16) has the same jumping indices α_k and nilpotency orders m_k , with only the coefficients $D_{km}(\xi^j \phi)$ depending on j .

Substituting into (6) and performing the Laplace integral term by term (substituting $u = n\beta t$ in each term), the j -th term contributes

$$\frac{(\sqrt{n}\beta)^j}{j!} \cdot \frac{D_{1,\theta}(\xi^j \phi) (\log n)^{\theta-1}}{(n\beta)^{\lambda+p/2+j/2}} \Gamma(\lambda + p/2 + j/2) = \frac{(\log n)^{\theta-1}}{(n\beta)^{\lambda+p/2}} \cdot \frac{\Gamma(\lambda + p/2 + j/2)}{j!} D_{1,\theta}(\xi^j \phi),$$

since $n^{j/2}/n^{\lambda+p/2+j/2} = n^{-(\lambda+p/2)}$, independently of j . The $\sqrt{n}$ scaling in the perturbation is exactly compensated by the half-integer shift of the Laplace exponent.

4.5 Restriction to the critical stratum

The sharp error bound requires the following factorization of the leading distribution through the critical stratum.

We work in normal crossing coordinates of a resolution $\pi: X' \rightarrow X$. After reordering, write the local coordinates as $(x, y) = (x_1, \dots, x_r, y_1, \dots, y_s)$ where

$$f \circ \pi = x^{2k} y^{2k'} = \prod_{i=1}^r x_i^{m_i} \prod_{j=1}^s y_j^{m'_j}$$

and the Jacobian is $|\det d\pi| = x^a y^{a'} = \prod x_i^{a_i} \prod y_j^{a'_j}$ (up to a smooth nonvanishing factor). The *critical variables* x are those achieving the minimum ratio: $(a_i + 1)/m_i = \lambda$ for $i = 1, \dots, r$. The *non-critical variables* y satisfy $(a'_j + 1)/m'_j > \lambda$ for $j = 1, \dots, s$. Define the *critical stratum* $S_\lambda = \{x = 0\}$, the *non-critical exponents* $\mu_j = a'_j - m'_j \lambda$, and the *gap* $\delta = \min_j \{(a'_j + 1)/m'_j - \lambda\} > 0$ (with $\delta = \min_i 1/m_i$ if there are no non-critical variables).

Lemma 26 (Restriction to the critical stratum). *In normal crossing coordinates:*

- (a) *The leading distribution satisfies $D_{1,\theta}(\psi) = \tilde{b} \int \psi(0, y) y^\mu dy$, where $\tilde{b}$ is the constant from Watanabe's Theorem 4.6. In particular, $D_{1,\theta}(\psi)$ depends on ψ only through $\psi|_{S_\lambda}$.*
- (b) *If ψ vanishes on S_λ , then $v_\psi(t) = O(t^{\lambda+\delta-1} |\log t|^N)$.*

Proof. By Taylor's theorem, $\psi(x, y) = \psi(0, y) + \sum_{i=1}^r x_i \psi_i(x, y)$ where $\psi_i(x, y) = \int_0^1 (\partial_{x_i} \psi)(sx, y) ds$ is smooth (and compactly supported if ψ is).

For (b), it suffices to show that $v_{x_i \psi_i}(t) = O(t^{\lambda+1/m_i-1} |\log t|^N)$ for each i . In the normal crossing integral

$$v_{x_i \psi_i}(t) = \int \delta(t - x^{2k} y^{2k'}) x^a y^{a'} x_i \psi_i(x, y) dx dy,$$

the extra factor x_i changes the effective exponent of x_i from a_i to $a_i + 1$. This integral has exactly the same form as the state density $v_{\psi_i}(t)$ for a function with Jacobian exponents $(a_1, \dots, a_i + 1, \dots, a_r, a'_1, \dots, a'_s)$. The minimum ratio for this modified integral is at least $\min(\lambda + 1/m_i, \lambda + \delta) = \lambda + \delta$ (since the i -th ratio increases from λ to $\lambda + 1/m_i \geq \lambda + \delta$, and all other ratios

are unchanged). By Watanabe's Theorem 4.6 applied to this modified integral, $v_{x_i\psi_i}(t) = O(t^{\lambda+\delta-1}|\log t|^N)$.

For (a), part (b) shows that the correction terms $x_i\psi_i$ are lower order, so the leading coefficient comes from $\psi(0, y)$ alone. The $\psi(0, y)$ contribution is computed by fixing y and integrating the critical variables using $\delta(t - x^{2k}y^{2k'}) = y^{-2|k'|}\delta(t/y^{2|k'|} - x^{2k})$ and the state density $w(u) = \tilde{b}u^{\lambda-1}(\log(b^{2|k|}/u))^{r-1}$ from Watanabe's Theorem 4.6. Collecting the powers $y^{a'_j - m'_j\lambda} = y^{\mu_j}$ and extracting the leading $(-\log t)^{\theta-1}$ power gives $D_{1,\theta}(\psi) = \tilde{b} \int \psi(0, y) y^\mu dy$. $\square$

4.6 The central coefficient

Summing the leading contributions from Section 7.4 over j , and using Lemma 26(a) to write $D_{1,\theta}(\xi^j\phi) = \tilde{b} \int \xi_0^j \phi_0 y^\mu dy$ (where $\xi_0 = \xi|_{S_\lambda}$, $\phi_0 = \phi|_{S_\lambda}$), we define the *central coefficient*

$$C_\xi^{(p)}(\phi) = \frac{\tilde{b}}{\beta^{\lambda+p/2}} \int_{[0,b]^s} \phi_0(y) y^\mu \int_0^\infty t^{\lambda+p/2-1} e^{-\beta t + \beta\sqrt{t}\xi_0(y)} dt dy. \quad (7)$$

The *central part* is the full leading term of the partition function:

$$Y^p(n, \xi, \phi) := \frac{(\log n)^{\theta-1}}{n^{\lambda+p/2}} \cdot C_\xi^{(p)}(\phi), \quad (8)$$

which corresponds to Watanabe's equation (4.19) in [6] (up to the identification of conventions for the exponent p).

Derivation of (7). Expanding $e^{\beta\sqrt{t}\xi_0}$ and integrating term by term:

$$\int_0^\infty t^{\lambda+p/2-1} e^{-\beta t + \beta\sqrt{t}\xi_0} dt = \sum_{j=0}^\infty \frac{\beta^j \xi_0^j}{j!} \int_0^\infty t^{\lambda+p/2+j/2-1} e^{-\beta t} dt = \sum_{j=0}^\infty \frac{\xi_0^j}{j!} \frac{\Gamma(\lambda + p/2 + j/2)}{\beta^{\lambda+p/2+j/2}}.$$

Multiplying by $\tilde{b}\phi_0 y^\mu$, integrating over y , and using $D_{1,\theta}(\xi^j\phi) = \tilde{b} \int \xi_0^j \phi_0 y^\mu dy$:

$$C_\xi^{(p)}(\phi) = \frac{1}{\beta^{\lambda+p/2}} \sum_{j=0}^\infty \frac{\Gamma(\lambda + p/2 + j/2)}{j!} D_{1,\theta}(\xi^j\phi).$$

This confirms that summing the leading Laplace contributions from Section 7.4 gives (7). $\square$

4.7 The sharp error bound

Theorem 27. *The partition function satisfies*

$$Z^p(n, \xi, \phi) = Y^p(n, \xi, \phi) + E^p(n, \xi, \phi), \quad (9)$$

where Y^p is the central part (8) and the error satisfies

$$|E^p(n, \xi, \phi)| \leq c_1 \frac{(\log n)^{\theta-2}}{n^{\lambda+p/2}} e^{\beta\|\xi\|^2/4} (\beta\|\nabla\xi\| \|\phi\| + \|\nabla\phi\| + \|\phi\|), \quad (10)$$

where $\|\nabla\xi\|$ and $\|\nabla\phi\|$ are measured in the normal crossing coordinates of a resolution.

Proof. The error $E^p = Z^p - Y^p$ decomposes into three sources:

$$E^p = \underbrace{E_{(a)}^p}_{\text{subleading logs}} + \underbrace{E_{(b)}^p}_{\text{higher poles}} + \underbrace{E_{(c)}^p}_{\text{off-stratum}}. \quad (11)$$

(a) *Subleading log powers.* For each j , the Laplace integral $\int_0^\infty e^{-n\beta t} t^{p/2+j/2} v_j(t) dt$ involves expanding $(\log(n\beta/u))^{\theta-1}$ after the substitution $u = n\beta t$. The terms with $(\log n)^{m-1}$ for $m < \theta$ contribute

$$E_{(a)}^p = O\left(\frac{(\log n)^{\theta-2}}{n^{\lambda+p/2}}\right) \sum_{j=0}^{\infty} \frac{\Gamma(\lambda + p/2 + j/2)}{j!} |D_{1,\theta}(\xi^j \phi)|.$$

Since $|D_{1,\theta}(\xi^j \phi)| \leq C \|\xi\|^j \|\phi\|$ and $\Gamma(\lambda + p/2 + j/2)/j!$ decays super-exponentially, the sum is bounded by $C' e^{\beta \|\xi\|^2/4} \|\phi\|$.

(b) *Higher poles.* The remainder $v_j(t) - \sum_{m=1}^{\theta} D_{1m}(\xi^j \phi) t^{\lambda-1} (-\log t)^{m-1}$ is $O(t^{\lambda_2-1} |\log t|^{N'})$ where $\lambda_2 > \lambda$ is the next pole. After the Laplace transform and $\sqrt{n}$ cancellation, this contributes $O(n^{-(\lambda_2+p/2)})$, which is $o(n^{-(\lambda+p/2)})$.

(c) *Off-stratum error.* The central coefficient $C_{\xi}^{(p)}(\phi)$ uses the on-stratum restrictions ξ_0, ϕ_0 , while the Laplace expansion in Section 7.4 uses the full $D_{1,\theta}(\xi^j \phi)$. By Lemma 26(a), $D_{1,\theta}(\xi^j \phi) = D_{1,\theta}((\xi_0)^j \phi_0)$, so $E_{(c)}^p = 0$: there is no off-stratum contribution at the leading log power.

However, the subleading distributions $D_{1,m}(\xi^j \phi)$ with $m < \theta$ are not necessarily supported on S_{λ} , so the bound in (a) involves $\|\xi^j \phi\|$ rather than $\|(\xi_0)^j \phi_0\|$. This is the origin of the $\|\nabla \xi\|$ and $\|\nabla \phi\|$ terms in (10): by the mean value theorem, $\|\xi^j \phi\|_{C^1} \leq C(j \|\xi\|^{j-1} \|\nabla \xi\| \|\phi\| + \|\xi\|^j \|\nabla \phi\| + \|\xi\|^j \|\phi\|)$, and the C^1 norm enters through the dependence of $D_{1,m}$ on derivatives of its argument.

Combining (a)–(c), the total error is bounded by (10). □

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